資料庫系統規格書

Fitness Tracking Management System

System Requirement Specification SRS

2026-09-23

v1.0

文件重點	說明
	PostgreSQL users exercises plans sessions sets body records
	PK FK M:N 3NF CRUD JOIN GROUP BY Aggregate Functions Cloud Database

Fitness Tracking Management System SQL
Dashboard
Primary Key Foreign Key

1

1.1

AI

Fitness Tracking Management System
Session

1.2

1.
2.
3.
4. Weight Reps Set Number
5. JOIN GROUP BY SUM COUNT SQL Dashboard
6. Exercise Database
7. PostgreSQL

1.3

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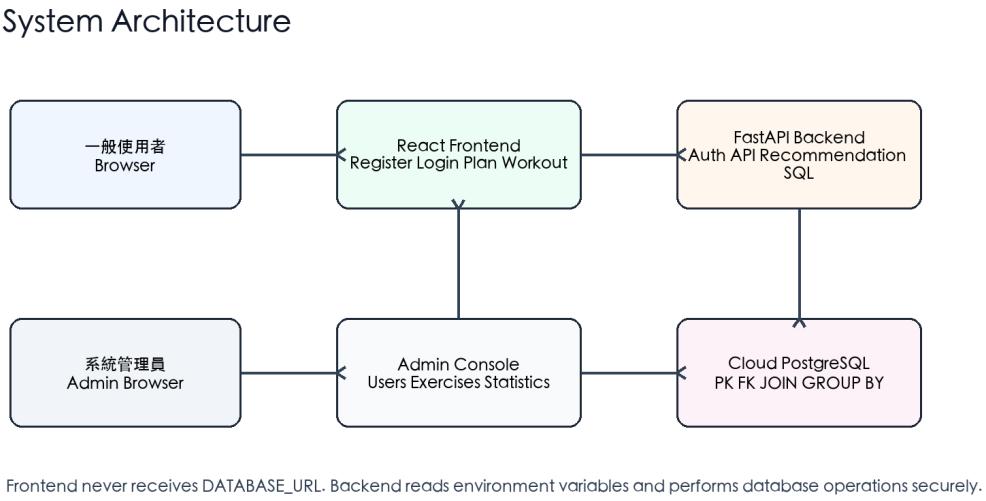
	Dashboard
	Exercise Database
	SQL

2

2.1



2.1

2.2

API Cloud PostgreSQL localhost database

DATABASE_URL SECRET_KEY API token

2.3

Layer		
Frontend	React TypeScript Vite	Dashboard
Backend	Python FastAPI	RESTful API
ORM	SQLAlchemy	Python model
Migration	Alembic	schema
Database	PostgreSQL	PK FK JOIN GROUP BY
Deployment	Render	Web Service Cloud PostgreSQL Demo localhost
Version Control	GitHub	migration README

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3.1

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Exercise

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Workout Plan

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Workout Session

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Workout Set

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Workout History Dashboard

3.2

Admin Login

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users workout statistics exercise database

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Exercise

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4

ID		
FR-01		users user_auth body_records user_body_parts token
FR-02		email/password last_login_at token
FR-03	Profile	
FR-04	Exercise Database	CRUD
FR-05		exercises workout_plans plan_days plan_exercises
FR-06		workout_sessions in_progress
FR-07		workout_sets set number
FR-08		set workout_sessions completed
FR-09		JOIN session exercise sets volume
FR-10	Dashboard	GROUP BY aggregate functions
FR-11	Admin	users exercises statistics

4.1 Auth Module

users user_auth body_records user_body_parts

rollback

- Name
- Email
- Password
- Gender
- Age
- Height
- Weight
- Training Experience
- Training Goal
- Training Days Per Week
- Training Duration
- Preferred Body Parts

Email

Email Email Email Email Email
@ user_auth Email

Email Email Email user_auth.email
NOT NULL UNIQUE

Email Email Email Service Token Token

Email		email
Email		user_auth.email NOT NULL
Email		user_auth.email UNIQUE
		email service token workflow
email_verified		

4.2 Profile Module

Profile Module users training_goals body_parts
user_body_parts body_records

4.3 Exercise Database Module

Exercise Database exercise
exercise

4.4 Workout Plan Module

exercises workout_plans plan_days plan_exercises

4.5 Start Workout Module

workout_sessions in_progress session_id workout session
workout_sets session completed

POST /users/{user_id}/workouts/sessions

Request: { plan_day_id }

Response: { session_id, status: "in_progress" }

POST /users/{user_id}/workouts/sessions/{session_id}/sets

Request: { exercise_id, set_number, weight_kg, reps }

Response: created workout_set

4.6 Workout History Dashboard

Workout History workout_sessions JOIN workout_sets JOIN exercises Dashboard GROUP BY Aggregate Functions Exercise

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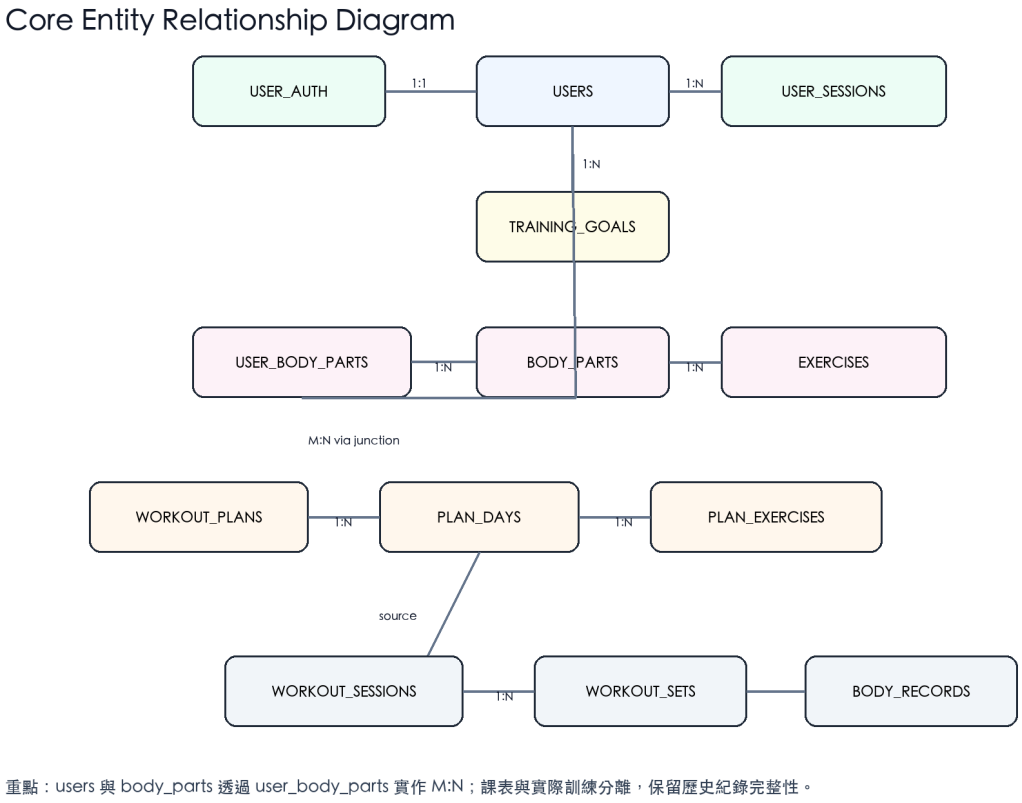
5.1

Table	
users	
user_auth	hash
user_sessions	session token
training_goals	reference table
body_parts	reference table
user_body_parts	users body_parts M:N junction table
exercises	
workout_plans	
plan_days	
plan_exercises	
workout_sessions	
workout_sets	weight/reps
body_records	
admins	

5.2 ER Diagram



5.1 ERD

5.3

Relationship	Type	
users - user_auth	1:1	
users - user_sessions	1:N	session
training_goals - users	1:N	
users - body_parts	M:N	user_body_parts
body_parts - exercises	1:N	
users - workout_plans	1:N	
workout_plans - plan_days	1:N	
plan_days - plan_exercises	1:N	
workout_sessions - workout_sets	1:N	
users - body_records	1:N	

6 Data Dictionary

6.1 users

			NULL	
user_id	UUID	PK	No	
name	VARCHAR(100)		No	
gender	VARCHAR(20)	CHECK	Yes	male female non_binary prefer_not_to_say
age	INT	CHECK	No	13 100
height_cm	DECIMAL(5,2)	CHECK	No	cm
training_experience	VARCHAR(20)	CHECK	No	beginner intermediate advanced
training_goal_id	INT	FK	No	training_goals
training_days_per_week	INT	CHECK	No	2 6
training_duration_minutes	INT	CHECK	No	30 45 60 90
is_active	BOOLEAN		No	
created_at	TIMESTAMP		No	
updated_at	TIMESTAMP		No	

6.2 user_auth

			NULL	
auth_id	BIGINT	PK	No	ID
user_id	UUID	FK UNIQUE	No	users.user_id user auth
email	VARCHAR(255)	UNIQUE NOT NULL	No	Email
password_hash	VARCHAR(255)		No	
email_verified	BOOLEAN		No	false
email_verified_at	TIMESTAMP		Yes	Email
last_login_at	TIMESTAMP		Yes	
created_at	TIMESTAMP		No	

6.3 user_sessions

			NULL	
session_id	BIGINT	PK	No	session ID

user_id	UUID	FK	No	users
token_hash	VARCHAR(255)		No	Token
issued_at	TIMESTAMP		No	
expires_at	TIMESTAMP		No	
revoked_at	TIMESTAMP		Yes	

6.4 training_goals

			NULL	
training_goal_id	INT	PK	No	ID
goal_code	VARCHAR(50)	UNIQUE	No	muscle_gain fat_loss strength general_fitness
goal_name	VARCHAR(100)		No	
description	TEXT		Yes	
default_sets	INT	CHECK	No	
default_reps	INT	CHECK	No	
default_rest_seconds	INT	CHECK	No	

6.5 body_parts

			NULL	
body_part_id	INT	PK	No	ID
body_part_code	VARCHAR(50)	UNIQUE	No	chest back legs
name_en	VARCHAR(50)	UNIQUE	No	
name_zh	VARCHAR(50)	UNIQUE	No	
display_order	INT	UNIQUE	No	
is_active	BOOLEAN		No	

6.6 user_body_parts

			NULL	
user_id	UUID	PK FK	No	users
body_part_id	INT	PK FK	No	body_parts
selected_at	TIMESTAMP		No	

6.7 exercises

			NULL	
exercise_id	BIGINT	PK	No	ID
exercise_name	VARCHAR(120)	UNIQUE	No	
body_part_id	INT	FK	No	
difficulty_level	VARCHAR(20)	CHECK	No	beginner intermediate advanced
equipment	VARCHAR(80)		No	
movement_type	VARCHAR(20)	CHECK	No	compound isolation
description	TEXT		No	
image_url	VARCHAR(255)		Yes	
instructions	JSON		Yes	
is_active	BOOLEAN		No	exercise

6.8 workout_plans

			NULL	
plan_id	BIGINT	PK	No	ID
user_id	UUID	FK	No	
training_goal_id	INT	FK	No	
plan_name	VARCHAR(120)		No	
training_days_per_week	INT	CHECK	No	
training_duration_minutes	INT	CHECK	No	
algorithm_version	VARCHAR(50)		No	rules-v1 llm-v1
status	VARCHAR(20)	CHECK INDEX	No	active archived user active plan
generated_at	TIMESTAMP		No	

6.9 plan_days

			NULL	
plan_day_id	BIGINT	PK	No	ID
plan_id	BIGINT	FK	No	
day_number	INT	UNIQUE(plan_id, day_number)	No	Day 1 Day 6

day_name	VARCHAR(100)		No	Chest + Triceps
focus_summary	VARCHAR(150)		No	

6.10 plan_exercises

			NULL	
plan_exercise_id	BIGINT	PK	No	ID
plan_day_id	BIGINT	FK	No	
exercise_id	BIGINT	FK	No	exercises
exercise_order	INT	UNIQUE(plan_day_id, exercise_order)	No	
target_sets	INT	CHECK	No	1 6
target_reps	INT	CHECK	No	1 30
rest_seconds	INT	CHECK	No	15 300
notes	TEXT		Yes	

6.11 workout_sessions

			NULL	
session_id	BIGINT	PK	No	session ID
user_id	UUID	FK	No	
plan_day_id	BIGINT	FK	Yes	
session_name	VARCHAR(120)		No	
status	VARCHAR(20)	CHECK	No	in_progress completed abandoned
started_at	TIMESTAMP		No	
completed_at	TIMESTAMP		Yes	
notes	TEXT		Yes	

6.12 workout_sets

			NULL	
workout_set_id	BIGINT	PK	No	Set ID
session_id	BIGINT	FK	No	workout session
exercise_id	BIGINT	FK	No	
set_number	INT	UNIQUE(session_id,	No	

		exercise_id, set_number)		
weight_kg	DECIMAL(6,2)	CHECK	No	kg
reps	INT	CHECK	No	
is_warmup	BOOLEAN		No	
logged_at	TIMESTAMP		No	
notes	TEXT		Yes	

6.13 body_records

			NULL	
body_record_id	BIGINT	PK	No	ID
user_id	UUID	FK	No	users
recorded_on	DATE	UNIQUE(user_id, recorded_on)	No	
weight_kg	DECIMAL(5,2)	CHECK	No	kg
notes	TEXT		Yes	
created_at	TIMESTAMP		No	

6.14 admins

			NULL	
admin_id	BIGINT	PK	No	ID
username	VARCHAR(80)	UNIQUE	No	
password_hash	VARCHAR(255)		No	
is_active	BOOLEAN		No	
last_login_at	TIMESTAMP		Yes	
created_at	TIMESTAMP		No	

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7.1 Primary Key Foreign Key

entity table Primary Key users.user_id exercises.exercise_id workout_plans.plan_id workout_sessions.session_id workout_sets.workout_set_id Foreign Key parent table workout_sets.session_id workout_sessions.session_id workout_sets.exercise_id exercises.exercise_id

7.2 Constraints

- user_auth.email NOT NULL UNIQUE
- user_body_parts composite primary key (user_id, body_part_id)
- workout_sets UNIQUE(session_id, exercise_id, set_number)
- training_experience difficulty_level movement_type status CHECK constraints
- body_records UNIQUE(user_id, recorded_on)

7.3 Indexes

- user_auth.email unique index
- exercises(body_part_id, difficulty_level, is_active) index
- workout_sessions(user_id, started_at) index
- workout_sets(session_id) index sets
- body_records(user_id, recorded_on) index

7.4 3NF

1NF body parts Chest Back Legs VARCHAR user_body_parts

2NF user_body_parts (user_id, body_part_id) selected_at

3NF workout_sets body_part_name workout_sets -> exercises -> body_parts

8 SQL

8.1

```
SELECT
    u.name,
    s.session_name,
    e.exercise_name,
    ws.set_number,
    ws.weight_kg,
    ws.reps,
    ws.weight_kg * ws.reps AS volume
FROM workout_sets ws
JOIN workout_sessions s ON ws.session_id = s.session_id
JOIN users u ON s.user_id = u.user_id
JOIN exercises e ON ws.exercise_id = e.exercise_id
WHERE u.user_id = :user_id
ORDER BY s.started_at DESC, ws.set_number;
```

8.2

```
SELECT
    SUM(ws.weight_kg * ws.reps) AS total_volume
FROM workout_sets ws
JOIN workout_sessions s ON ws.session_id = s.session_id
WHERE s.user_id = :user_id
  AND s.status = 'completed'
  AND ws.is_warmup = FALSE;
```

8.3

```
SELECT
    bp.name_en,
    bp.name_zh,
    COUNT(*) AS set_count
FROM workout_sets ws
JOIN exercises e ON ws.exercise_id = e.exercise_id
JOIN body_parts bp ON e.body_part_id = bp.body_part_id
JOIN workout_sessions s ON ws.session_id = s.session_id
WHERE s.user_id = :user_id
  AND s.status = 'completed'
GROUP BY bp.name_en, bp.name_zh
ORDER BY set_count DESC;
```

8.4 Exercise

```
SELECT
    e.exercise_name,
    SUM(ws.weight_kg * ws.reps) AS total_volume
FROM workout_sets ws
JOIN exercises e ON ws.exercise_id = e.exercise_id
JOIN workout_sessions s ON ws.session_id = s.session_id
WHERE s.user_id = :user_id
  AND s.status = 'completed'
GROUP BY e.exercise_name
ORDER BY total_volume DESC;
```

8.5

```
SELECT
    recorded_on,
    weight_kg
FROM body_records
```

WHERE user_id = :user_id

ORDER BY recorded_on ASC;

9 Workout Recommendation Algorithm

rule-based algorithm AI API LLM exercises

8. profile training_goal experience days duration
9. user_body_parts
10. exercises active body part difficulty
11. training_goal sets reps rest_seconds
12. days per week plan_days
13. preferred body parts
14. workout_plans plan_days plan_exercises

9.1 Start Workout

-> INSERT INTO workout_sessions(user_id, plan_day_id, session_name, status, started_at)

-> status = in_progress

-> session_id

->

-> INSERT INTO workout_sets(session_id, exercise_id, set_number, weight_kg, reps)

->

-> UPDATE workout_sessions SET status = 'completed', completed_at = NOW()

10

	API 95% 1000ms Dashboard 95% 1500ms
	Password token
	DATABASE_URL SECRET_KEY API Key commit GitHub Environment Variables
	Chrome Safari Edge
	Cloud PostgreSQL localhost
	Alembic migration schema README .env.example

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TC-01		Email	Email Password	users user_auth body_records user_body_parts Token
TC-02	Email	Email	Email	
TC-03			Email Password	Token
TC-04			Password	
TC-05		Profile		workout_plans plan_days plan_exercises
TC-06		active plan		workout_sessions status in_progress
TC-07	Set	workout session	Weight/Reps	workout_sets
TC-08		workout_set		workout_sessions.status completed
TC-09	Dashboard		Dashboard	Exercise
TC-10	API	Token	API	401 Unauthorized
TC-11	Admin Exercise	Admin	Exercise	exercises
TC-12		A	B	

12

DATABASE_URL	PostgreSQL .env
SECRET_KEY	Token GitHub
ADMIN_BOOTSTRAP_USERNAME	
ADMIN_BOOTSTRAP_PASSWORD	
OPENAI_API_KEY	AI rule-based algorithm
OPENAI_MODEL	AI
FRONTEND_URL	
CORS_ORIGINS	

Source Code GitHub Web Service Cloud PostgreSQL Render
GitHub repository

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15. users user_auth
16. user_body_parts User Body Part M:N

17. workout_plans plan_days plan_exercises
18. workout_sessions workout_sets
19. body_records
20. PK FK UNIQUE CHECK constraints
21. JOIN GROUP BY SUM COUNT SQL Dashboard
22. Cloud PostgreSQL localhost

14

Fitness Tracking Management System Exercise Database
Dashboard

Primary Key Foreign Key M:N junction table JOIN Aggregate Functions